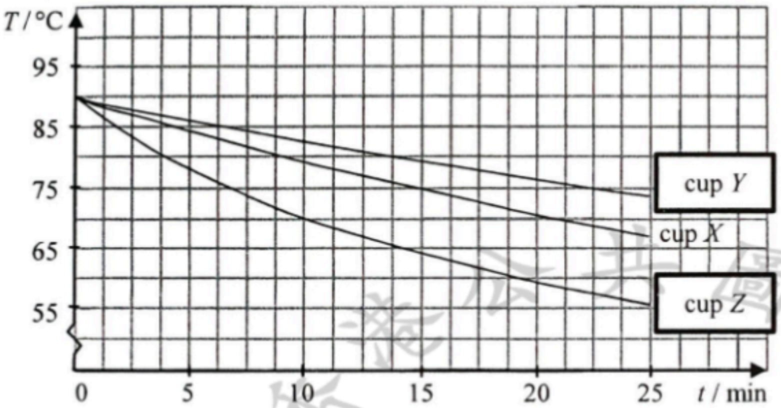
1.	(a)	Fair test (otherwise the initial temperature would affect the result).	1A
			1
	(b)	Temperature difference between hot water and the surroundings (room temperature) decreases as time elapses.	1A
		Rate of energy loss / heat loss becomes smaller, hence rate of temperature drop is smaller and the curve becomes less steep.	1A
			2
	(c)	(i) The graphs of T against t	1A
			1

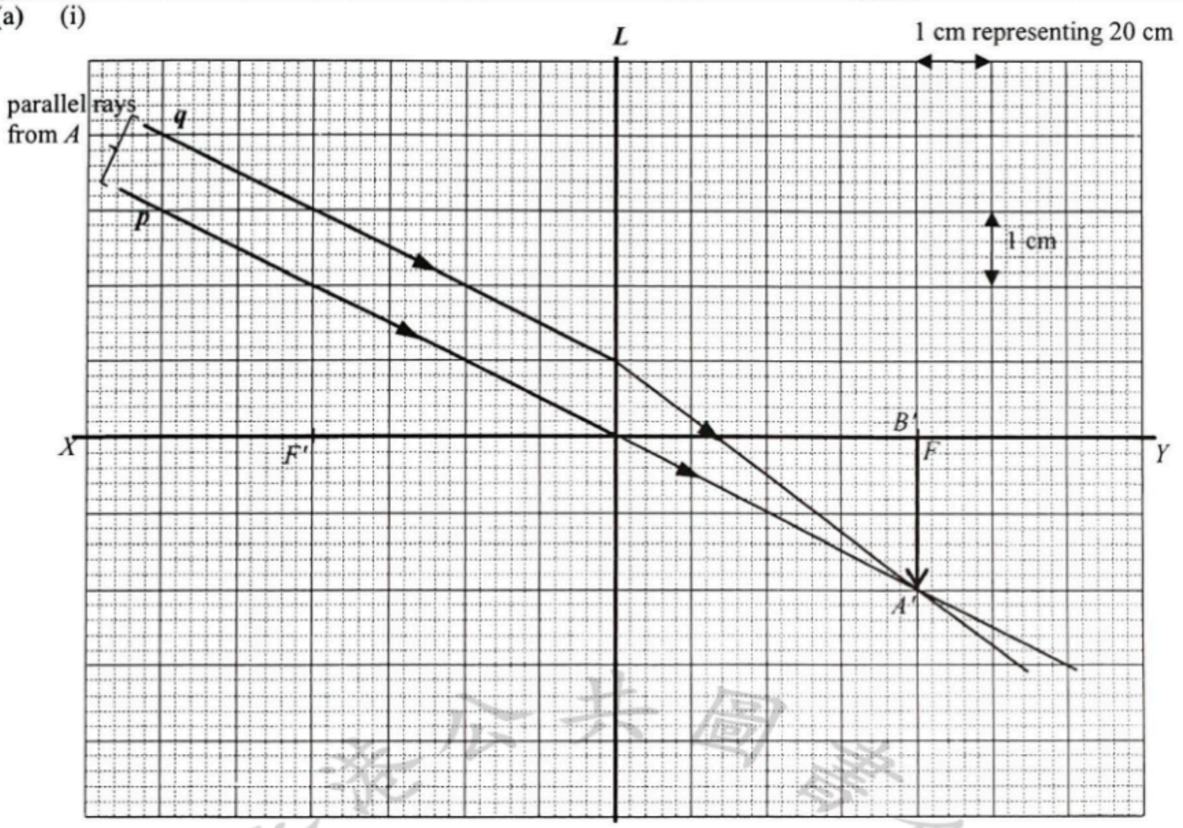


		(ii) Comparing to cup X as a control,	1A
		Cup Y: The shiny surface of aluminium foil is a poor emitter / good reflector of radiation (thus reduces heat lost to surroundings).	1A
		Cup Z: No lid allows convection through air / surroundings.	1A
		Or No lid enhances heat lost by evaporation / convection / contacting with air / surroundings.	1A
			3
	(d)	Polystyrene / foam plastic or foam / wood	1A
			1

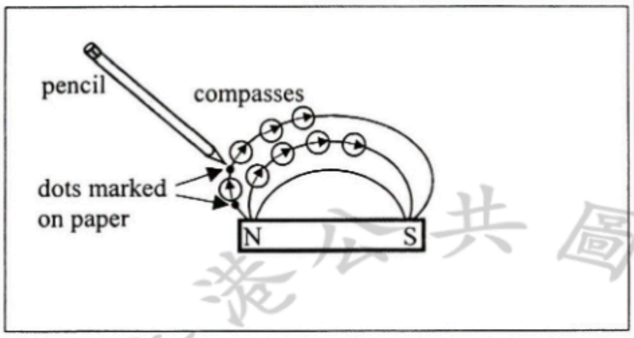
2. (a) (i) (I) $pV = nRT$ $(1.0 \times 10^5)(6.0 \times 10^{-4}) = n(8.31)(300)$ $n = 0.0240674$ number of molecules $N = nN_A$ $= (0.0240674)(6.02 \times 10^{23})$ $= 1.448857 \times 10^{22} \approx 1.45 \times 10^{22}$	1M 1A	
(II) average kinetic energy of gas molecules $E_K = \frac{3}{2}\left(\frac{R}{N_A}\right)T$ $= \frac{3}{2}\left(\frac{8.31}{6.02 \times 10^{23}}\right)(300)$ $= 6.211794 \times 10^{-21}\ \text{J} \approx 6.21 \times 10^{-21}\ \text{J}$	 1M 1A	
	4	
(ii) (I) N and E_K remain unchanged.	1A+1A	
	2	
(II) $E_K = \frac{1}{2}mc_{r.m.s.}^2$ $\frac{1}{2}m(600)^2 = \frac{1}{2}(\frac{1}{5}m)c_{r.m.s.}^2$ $\Rightarrow c_{r.m.s.} = \sqrt{5} \times 600$ $= 1341.6408\ \text{m s}^{-1} \approx 1340\ \text{m s}^{-1}$	 1M 1A	
	2	
(b) As gas C diffuses into the upper jar, its molecules collide with the air molecules and thus describe a zig-zag path / not along a straight path.	1A 1A	
	2	

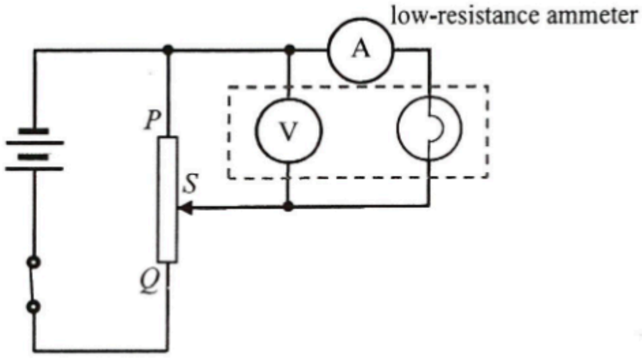
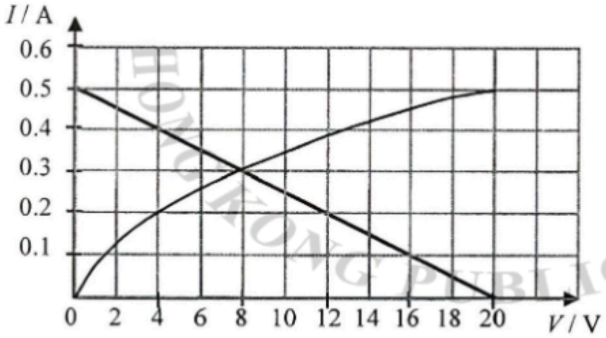
3. (a) According to Newton's third law of motion, force on air streams and the thrust on the quadcopter are (equal and) opposite, with magnitude given by F = rate of change of momentum (of air streams).	1A	
Therefore the thrust (upward) on the quadcopter balances its weight for a certain speed of air streams.	1A	
	2	
(b) (i) Volume of air streams propelled in 1 s $V = 0.284\ v$	1M	
m_a = density × volume $= 1.20 \times 0.284\ v = 0.3408\ v$	1A	
	2	
(ii) Weight of quadcopter = rate of change of momentum of air streams $1.38\ g = m_a v - 0$	1M	
$1.38 \times 9.81 = (0.3408\ v) \times v$ (From (b)(i)) $v^2 = 39.723592$ $v = 6.302665\ \text{m s}^{-1} \approx 6.30\ \text{m s}^{-1}$	1A	For $g = 10\ \text{m s}^{-2}$, $v = 6.363408\ \text{m s}^{-1} \approx 6.36\ \text{m s}^{-1}$
	2	
(c) (i) 	2A	
	2	
(ii) centripetal force required $F = \frac{1.38 \times 15^2}{50}$	1M	
$= 6.21$ N	1A	
	2	
(iii) Let T = total thrust on the quadcopter Vertical: $T \cos \theta = 1.38\ g$ ($g = 9.81\ \text{m s}^{-2}$) Horizontal: $T \sin \theta = 6.21$ N (From (c)(ii))	1M	Accept $\tan \theta = \frac{\text{centripetal force}}{\text{weight}}$
On solving the two equations, $\tan \theta = 0.458716$ $\theta = 24.641662° \approx 24.6°$	1A	For $g = 10\ \text{m s}^{-2}$, $\tan \theta = 0.45$ $\theta = 24.227745° \approx 24.2°$
	2	

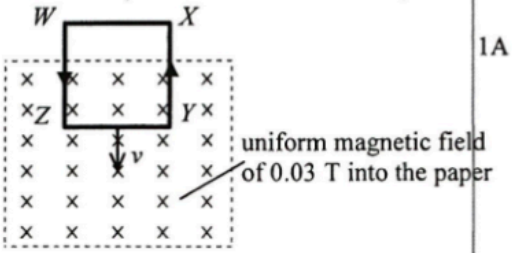
4.	(a)		$\frac{1}{2}mv^2 = mgh$ $= (50)(9.81)(1.5)$ $= 735.75\text{ J} \approx 736\text{ J}$	1M 1A	For $g = 10\text{ m s}^{-2}$, K.E. = 750 J
				2	
	(b)	(i)	Kinetic energy and potential energy (of the athlete) change to elastic potential energy (of the trampoline).	2A	
				2	
		(ii)	$\bar{F}d = mgh + mgd$ [Or $\bar{F}d = \frac{1}{2}mv^2 + mgd$] $\bar{F} = \frac{50(9.81)(1.5+0.40)}{0.40}$ $= 1839.375 + 490.5$ $= 2329.875\text{ N} \approx 2330\text{ N}$	1M 1A	For $g = 10\text{ m s}^{-2}$, $\bar{F} = 2375$ N
			Or $\bar{F} = \frac{1}{0.40}[735.75 + (50)(9.81)(0.40)]$ ≈ 2330 N	1M 1A	
				2	

Solution	Marks	Remarks
5. (a) (i)		
		
refracted rays of p, q correctly drawn and	1A	
A' correctly marked.	1A	
image $A'B'$ correctly marked.	1A	
	3	
(ii) Image $A'B'$ of a distant object, say, a building is real and therefore it can be captured	1A	
by a screen (placed at the focal plane).	1A	
	2	
(b) (i) Ratio by similar triangles, $\dfrac{\text{height of } AB}{\text{object distance}} = \dfrac{\text{height of } A'B'}{\text{focal length /image distance}}$		
$= \dfrac{2}{4\times 20} = \dfrac{1}{40}$	1M	Accept height of image $A'B'$: 1.8 cm ~ 2.2 cm
$= 0.025$	1A	
	2	
(ii) $\dfrac{\text{height of } AB}{\text{object distance}} = \dfrac{1}{40} = 0.025$		
Height of $AB = 0.025 \times 200 = 5$ m	1A	
	1	

Solution	Marks	Remarks
6. (a) Sound waves having the same frequency / wavelength and a constant phase difference (or always in phase / in opposite phase) are coherent.	1A	
	1	
(b) (i) When the sound waves from A and B meet at various positions along OY, interference occurs.		
At positions where the two waves are in phase, constructive interference occurs and gives maximum (loudness).	1A	
At positions where the two waves are in opposite phase, destructive interference occurs and gives minimum (loudness).	1A	
	2	
(ii) Any ONE: • due to background noise • due to reflection of unwanted sound from the wall, floor etc. • the intensity of sound waves from A and B reaching P are NOT equal as $AP < BP$ therefore cancellation is incomplete.	1A	
	1	
(c) Path difference at Q $3\lambda/2 = 2.58 - 2.17$ $= 0.41$ m $\therefore\ \lambda = 0.27333333 \text{ m} \approx 0.273$ m	1M	
$v = f\lambda$ $= (1200)(0.273)$ $= 328 \text{ m s}^{-1}$	1A	Accept 327.6 m s^{-1} to 328 m s^{-1}
	2	
(d) Path difference Δ at any position along OY from A and B is less than AB (i.e. $\Delta = n\lambda < AB$) max. $\Delta = 0.80 \text{ m} = \frac{0.8}{0.273}\lambda \approx 2.93\lambda$ i.e. max. $\Delta < 3\lambda$	1M	
As path difference cannot be equal to $3\lambda, 4\lambda, \cdots$, therefore no more maximum beyond R.	1A	
	2	
(e) increases	1A	
	1	

7.	(a)	(i)	*A*: south pole (S) *B*: south pole (S)	1A	
				1	
		(ii)	(Towards) bottom / downwards	1A	
				1	
		(iii)	Iron filings come into physical contact with the magnet will never come off / are difficult to get rid of.	1A	
				1	
	(b)	(i)	Successively place the compasses on the paper near one pole (say N) of the magnet such that the tip of each compass needle follows / lines up with the tail of the compass ahead.	1A	
			Diagram	1A	
					
			Denote the tip and tail (of each compass) with dots on the paper using the pencil <u>or</u> remove the compasses one by one and trace the (direction of) field line.	1A	
			Draw / sketch a smooth curve representing a field line by joining up the series of dots/segments going from one pole to the other.	1A	
			Repeat the above process by starting from different points around the magnet to get another/several field line(s).	1A	
				5	
		(ii)	Any ONE: • Compass is sensitive enough to explore very weak magnetic field such as the Earth's field. • The drawing can show the direction of the field lines / the drawing of the field will still be on the paper after removing the magnet. • It is difficult to draw individual field line separately in iron-filing method even if the filings are sprinkled very thinly and evenly.	1A	
				1	

Solution	Marks	Remarks
8. (a) (i) 	1A	
	1	
(ii) Brightness increases.	1A	
	1	
(b) (i) Resistance = $\frac{V}{I} = \frac{20}{0.5}$	1M	
$= 40\ \Omega$	1A	
	2	
(ii) As the applied voltage V increases, current I / electrical power increases	1A	
and this raises the filament's temperature,	1A	
thus the resistance R of the filament / bulb increases.		
Or the electrons collide with the increasingly vibrating atoms / lattice ions (of the filament) which impede their flow, i.e. R increases.	1A	
	2	
(c) (i) 	1M	Correct straight line intersects the curve at (8, 0.3)
Current = 0.3 A (at V = 8 V)	1A	
	2	
(ii) Power consumed = IV		
$= (0.3)(8)$	1M	e.c.f. from (c)(i)
$= 2.4$ W	1A	
	2	

9. (a) Induced current is anticlockwise (i.e. in the direction $ZYXW$)	1A		
		1	
(b) Current $I = \frac{\varepsilon}{R} = \frac{Blv}{R}$ [Or $\varepsilon = \frac{N\Delta\Phi}{\Delta t} = \frac{B\Delta A}{\Delta t} = Blv = IR$]			
$0.01 = \frac{0.03(0.10)v}{4\times0.15}$	1M		
$v = 2.0\ \mathrm{m\ s^{-1}}$	1A		
Or Power input $P = Fv = I^2R$ $(IlB)v = I^2R$ $Blv = IR$			
$(0.03)(0.10)\ v = (0.01)(4 \times 0.15)$	1M		
$v = 2.0\ \mathrm{m\ s^{-1}}$	1A	2	
(c) (i) $V_{YZ} = I\,(R_{ZW} + R_{WX} + R_{XY})$	1M		
$= 0.01(0.15 + 0.15 + 0.15)$			
$= 0.0045\ \mathrm{V}\ (= 4.5\ \mathrm{mV})$	1A		
Or $V_{YZ} = \varepsilon - IR_{YZ}$	1M		
$= Blv - IR_{YZ}$ $= 0.03(0.10)(2.0) - 0.01(0.15)$			
$= 0.0045\ \mathrm{V}$	1A	2	
(ii) As there is voltage drop (IR_{YZ}) within / across YZ, V_{YZ} is less than the induced e.m.f. across YZ ($\varepsilon = Blv$)	1A	1	Note: $\varepsilon - IR_{YZ} = V_{YZ}$ $\varepsilon = 0.006\ \mathrm{V}$

10.	(a)	Decrease in mass in the reaction (for two ${}^{2}_{1}H$) $\Delta m = (2 \times 2.014102) - (3.016029 + 1.008665)$ $= 4.028204 - 4.024694 = 0.003510$ u Maximum energy released by 1 mole of hydrogen nuclides $= \left(\frac{6.02 \times 10^{23}}{6420 \times 2}\right) \times (0.003510 \times 931 \text{ MeV})$ $= 1.532104 \times 10^{20}$ (MeV) $\approx 1.53 \times 10^{20}$ (MeV)	 1M 1M 1A	3
	(b)	${}^{2}_{1}H + {}^{2}_{1}H \rightarrow \underline{{}^{3}_{1}H + {}^{1}_{1}H}$ (or $\underline{{}^{1}_{1}p}$)	1A	1
	(c)	Any TWO: - Unlike fission, the fuel (hydrogen) for fusion is abundant and widely available. - Unlike fission, the products of fusion reactions are not radioactive. - Much larger amount of energy is produced by fusion (for equal mass of fuel).	2A	2